RideAware
See Beyond

Feasibility Report

For the Financial Year Ended
31 December 2027 (FY2027)

Our Idea

Intelligent photochromic cycling glasses that enhance navigation and situational awareness.

NEW ENTERPRISE DEVELOPMENT PROJECT

Module Code	BAA1003
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DCU

Value Design Proposition

Customer Jobs

- Use lightweight, comfortable, easy-to-handle tech
- Integrate with existing apps (Komoot, Strava, Maps, Garmin)
- Ensure long battery life and stable connectivity
- Avoid stopping to check a phone or bike computer
- Follow clear, reliable, real-time directions
- Reduce stress in traffic and unfamiliar areas
- Keep eyes on the road at all times
- Navigate safely without looking down
- Signal sustainable and responsible mobility behaviour
- Find efficient routes in urban and mixed environments
- Present themselves as sporty, modern, and tech-savvy
- Feel confident and relaxed while riding

Gains

- More confident, enjoyable, stress-free rides
- Lightweight, comfortable, ergonomic design
- Unobstructed vision with minimal distraction
- Higher safety and situational awareness
- Fewer accessories needed (all-in-one solution)
- Seamless app and device integration
- Clear AR navigation in the line of sight

Pains

- Distraction from phones or bike computers
- Missed turns and unsafe situations
- Poor visibility (rain, glare, low light)
- Unclear or inconsistent cycling infrastructure
- Short battery life and unreliable performance
- Devices are bulky, complex, or expensive
- Audio-only navigation hard to hear

Gain Creators

- Removes need for unsafe phone-based navigation
- Ensures audible cues even in noisy traffic (bone conduction)
- Provides clear, distraction-free guidance in the field of view
- Replaces multiple accessories (sunglasses, navigation device)
- Improves visibility and usability in rain, glare, and low light
- Reduces technological burden through easy setup and familiar app integration
- Offers a simpler, more affordable alternative to complex AR glasses

Products & Services

- Bone-conduction audio for safe, ears-open guidance
- Weather-resistant hardware for rain, wind, and cold
- AR cycling glasses with real-time turn-by-turn navigation
- Photochromic, wind- and sun-protective lenses
- Lightweight, ergonomic design for commuting & sport
- App integration with reliable routing and familiar navigation tools
- Visual cues projected into the rider's line of sight

Pain Relievers

- Simple, intuitive interface increases user confidence and adoption
- Combines navigation + sun protection + wind protection in one device
- Affordable, accessible smart-glasses option for everyday cyclists
- All-weather usability increases comfort and ride consistency
- Enhances rider confidence through eyes-forward navigation
- Lightweight, ergonomic design supports long-distance comfort
- Reduces mental load and supports smoother, safer riding

RideAware©

Value Design Proposition

This section uses the Value Proposition Canvas (VPC) framework to identify and analyse key tasks, issues and benefits for our target group: cyclists and micro-mobility users. We have aligned our value map based on these insights, comprising product features, pain relievers and gain creators, with our customer profile. This ensures that we develop a user-centric and compelling product offering (Osterwalder, 2014).

Methodology

To evaluate product-market fit and define a feasible value proposition, we employ a mixed-methods approach inspired by Mooi and Sarstedt (2011). Our primary research consisted of a qualitative phase involving seven in-depth interviews with potential users, as well as expert consultations to test the technical feasibility of our product concept. Quantitative research is planned as a follow-up to validate key assumptions (see Appendix 1). This staged method aims to extract customer motivations, needs, and concerns relating to riding and navigation, and refine both user and technical feedback.

Primary research

We conducted seven semi-structured interviews with individuals aged between 20 and 60 who regularly cycle for commuting, leisure, sport, and urban navigation. Interviewees were selected for diversity in age, lifestyle, and riding behaviours. In addition, some cycle in dense urban environments such as Dublin and major German cities, while others ride on mountain trails or in rural areas, including road cyclists, recreational riders, and daily commuters.

Each interview lasted approximately 20 minutes and followed a set of nine guiding questions covering navigation habits, safety concerns, product reactions, design expectations, and willingness to pay.
At the core, the purpose of these questions was to understand how cyclists currently navigate, when and why they feel unsafe, how they perceive AR glasses as a concept, what features they value, and whether they would consider adopting such a product. These qualitative insights helped us refine both the product concept and its value proposition.

In parallel, we conducted a technical feasibility assessment through an expert survey distributed to researchers affiliated with the Sensor Technology Department at DCU, as well as conducting one in-depth expert interview. Consultations with three additional specialists in human–computer interaction, embedded systems, computing, and AR/VR helped us to identify weaknesses in our initial concept model and highlight key risks requiring redesign.

Secondary Research

To complement our qualitative findings, we conducted secondary research using both academic literature and industry databases. This included recent publications on smart wearables, augmented reality (AR) technologies, and urban mobility trends. Key areas of focus included AR display performance in outdoor conditions, user adoption of navigation tools, and the growing demand for lightweight, hands-free rider accessories.

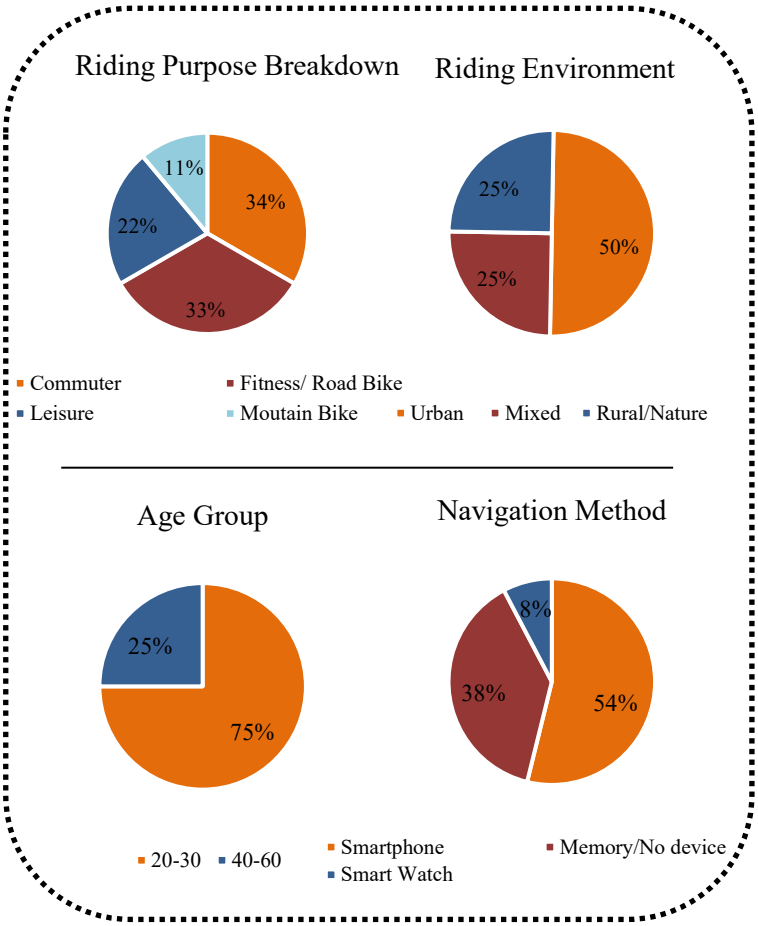


Figure 1: Summary of Interview Participant Demographics and Riding Behaviour.

Value Design Proposition

Customer Jobs

To briefly explain, customer jobs refer to the functional, social, and emotional tasks users aim to accomplish when cycling or using other micro mobility tools to navigate in daily life or sporting contexts (Osterwalder, 2014). Specific to our product offering, these jobs can be divided into functional, social, and emotional dimensions.

1. Functional Jobs

All interviewed participants emphasised the need for reliable, timely and safe navigation while cycling, without having to look down at a smartphone or a bike computer. Riders want to maintain focus on the road, avoid stopping to orient themselves, and find optimal routes in urban environments with inconsistent cycling infrastructure or during long-distance sport rides. For commuters, time efficiency and safe arrival are priorities, while performance cyclists value training optimisation and improved aerodynamics through hands-free navigation. Additionally, they value lightweight, easy-to-handle wearable technology with long battery life that seamlessly connects to existing devices such as smartphones and the app.

2. Social Jobs

Cyclists indicated a strong desire to present themselves as sporty, modern, and health-conscious individuals who pursue sustainable mobility choices. Using advanced cycling technology allows them to look stylish and demonstrate responsible road behaviour. Many also seek to signal alignment with environmental and active-lifestyle values and support a broader societal transition toward greener transportation modes.

3. Emotional Jobs

Cyclists want to feel safe, confident, and relaxed when riding, especially in stressful traffic conditions or unfamiliar areas. Riders expressed a strong desire to reduce anxiety caused by unclear road design and navigation errors. There is also emotional value in knowing they are supported by a tool which improves riding enjoyment. Sport cyclists associate emotional satisfaction with maintaining rhythm, focus, and flow without visual interruption.

Customer Pains

Customer pains represent barriers, risks, or frustrations encountered while trying to achieve safe cycling navigation (Osterwalder, 2014).

1. Safety & Distraction Pains

Many riders feel unsafe whenever they look away from the road to check a phone or bike computer. Screen-based or audio-only navigation often causes missed turns or risky moments in traffic. Interviewees stressed that any AR solution must not obstruct their field of view, and that rain, glare or low light further increase perceived danger.

2. Navigation and infrastructure Pains

Unclear or inconsistent cycling infrastructure makes navigation stressful, especially at junctions or when routes change suddenly. Existing tools offer too little situational awareness, leaving riders feeling uncertain in unfamiliar or complex environments.

3. Technology & Product-Related Pains

Current smart glasses are seen as expensive, complex and hard to use. Riders criticised short battery life, unintuitive interfaces and the need to carry multiple accessories. Interviewees noted that there is no multifunctional solution that addresses these needs in a simple, reliable way.

Customer Gains

Customer gains describe the positive outcomes and benefits cyclists aim to achieve with improved navigation technology (Osterwalder, 2014).

1. Clear Navigation and Multifunctional glasses

Cyclists want clear, timely and unobtrusive navigation. They expect turn-by-turn guidance that preserves their full field of vision and enables them to keep their eyes on the road at all times. Reliable operation, long battery life and a lightweight, ergonomic design were repeatedly cited as essential requirements. Seamless integration with existing apps and devices is considered a basic expectation, while multifunctional features, such as wind, sun and weather protection, add further value.

2. Safety, Confidence and Relaxed Riding

Emotionally, riders aim to feel safer, calmer and more confident during their journeys. Interviewees described how current navigation methods often cause stress, especially in unfamiliar areas or heavy traffic. A distraction-free system that supports continuous forward attention reduces mental load and helps cyclists maintain focus and flow. The increased sense of reassurance and comfort contributes to a more enjoyable riding experience.

3. Lifestyle, Identity & Social Value

Cyclists value technologies that reflect their identity and lifestyle. Many appreciate modern, minimalist equipment that signals sportiness, health consciousness and sustainable mobility. Discreet, innovative eyewear reinforces their self-image as responsible and forward-thinking riders.

4. Mobility Freedom & Everyday Efficiency

Several interviewees also highlighted increased mobility as a key benefit. Clear, efficient navigation helps them travel faster, avoid errors and rely less on cars or public transport. Greater confidence in navigating complex or unfamiliar routes reinforces cycling as a practical, everyday mode of transport.

Customer Profile

Segmentation

To identify RideAware's most relevant audience, we applied a multi-dimensional segmentation approach including psychographic, geographic, behavioural, and socio-demographic dimensions. Interview insights show that customers value efficiency, safety, sports performance, and practical mobility solutions while seeking products that align with a healthy and eco-conscious lifestyle.
Geographic segmentation focused on cyclists in urban environments with poor cycling infrastructure, high traffic density and navigation complexity. Behavioural segmentation explored cycling frequency, typical usage context, willingness to pay, reaction to pain points. Socio-demographic segmentation considered variables such as age, income, occupation, and education.

Quantification

Primary research from interviews revealed several dominant patterns that shaped prioritisation of market segments:

High relevance of safety improvement and stress reduction in traffic, especially for older cyclists and urban riders. Navigation support is significantly more valued by younger cyclists, who often use their phone for route guidance and experience distraction and frustration. Strong interest in products that combine functions (photochromic or prescription-ready lenses, sunglasses replacement, wind protection).

Price tolerance centred around €100–€250, indicating a willingness to adopt smart cycling eyewear if the product offers clear functional value rather than unnecessary premium technology features.
Cycling for sports, leisure, exploring new routes or navigating unfamiliar cities showed significantly higher purchase intent compared with daily commuting only.

Quantification confirmed that the highest-potential early adopter market consists of tech-interested active cyclists looking for efficiency, safety and convenience.

Clustering

Based on segmentation and research insights, we defined three core customer clusters:

1. Urban Rider and Delivery Drivers

Values safety, clarity and confidence in complex traffic environments. Focuses on accident prevention and reducing stress through navigation.
Often older or middle-aged cyclists, commuting or riding in dense urban areas; or delivery drivers cycling new routes daily.
Highly attracted to features such as real-time navigation, visibility in field of view and audio prompts.

2. Performance-Focused Explorer

Sporty cyclist using cycling for fitness, adventure or performance improvement. Frequently rides unfamiliar routes for training, competition or leisure. Values accuracy, minimal distraction, long battery life, aerodynamic design.

Value Design Proposition

3. Tech-Savvy Mobility Optimiser

Pragmatic modern users see cycling as an efficient and sustainable transport alternative. Younger demographic, urban lifestyle, high smartphone reliance. Motivated by convenience, sustainability, and multifunctional design. Seeks products that save time, reduce hassle and look stylish.

These clusters reflect behavioural patterns such as struggling with phone-based navigation, feeling unsafe in traffic, or wanting multifunctional eyewear that replaces existing sunglasses.

Application

The personas guided strategic decisions related to product development, pricing, and go-to-market execution. Our Safety Positioning highlights accident-reduction potential, eyes-forward navigation, enhanced focus on surroundings and targets the Safety-Driven Urban Rider.

Sport & Performance use case emphasises seamless route guidance, aerodynamic design, sport aesthetic, long battery life, weather resistance and Targets the Performance-Focused Explorer.

Lifestyle & Convenience communicates integration with favourite cycling apps, replace-your-sunglasses functionality, photochromic lenses and tech usability and Targets the Tech-Savvy Mobility Optimiser.

Products & Services

This section highlights the core offering RideAware provides to address customer needs. RideAware offers AR cycling glasses that display real-time navigation and visual guidance directly in the cyclist's field of view.

The glasses allow riders to maintain full focus on the road and surrounding traffic while navigating unfamiliar routes. Designed for sport, commuting, and urban mobility, RideAware combines navigation functionality with high-quality protective eyewear, replacing the need for conventional sunglasses or frequent phone checks.

The glasses integrate with our app providing reliable route guidance. The lightweight, ergonomic design includes photochromic lenses, wind and sun protection, and optional prescription compatibility, ensuring everyday usability beyond cycling. With a battery life designed to last throughout long rides and weather-resistant hardware suitable for demanding conditions, RideAware supports a seamless cycling experience across varied environments, including cities with poor cycling infrastructure and rural sport routes.

Pain Relievers

Pain relievers are solutions RideAware provides to directly address the frustrations and challenges cyclists experience (Osterwalder, 2014).

1. Clear, Distraction-Free Navigation

Eliminates the need for unsafe and distracting phone-based navigation by placing guidance directly in the user's field of view and through bone-conduction technology audio prompts. Therefore, reduces stress and frustration associated with navigating poorly marked or inconsistent cycling infrastructure. This results in enhanced personal safety perception, especially for riders navigating busy or unfamiliar areas. Moreover, it solves the problem of difficult-to-hear audio instructions in noisy traffic environments.

2. Multifunctional Glasses

Offers weather-resistant hardware that remains usable in rain, wind, sun, and cold, unlike smartphones, which are harder to use safely in poor weather. It replaces the need for separate sunglasses, resolving the pain of carrying multiple accessories.

3. Overcrowded Costly Smart glasses market

It provides a lower-cost alternative to overly complex and expensive AR glasses, focusing on essential functions only. Moreover, it reduces technical burden through simple setup, familiar app integration, and intuitive interaction

"I'd actually use smart glasses if they were simple and not too expensive." **Joel Helber**

Value Design Proposition

Gain Creators

Gain creators describe how RideAware enables customers to achieve desired outcomes and enhance their cycling experience (Osterwalder, 2014).

1. Clear, Distraction-Free Navigation

Real-time turn-by-turn navigation projected directly into the rider's line of sight ensures continuous focus on the road. Visual cues will be projected on the peripheries of a user's field of view and will be unintrusive to ensure minimal distraction. As a result, riders will experience greater confidence and reduced mental load when cycling in unfamiliar or high-traffic environments.

Audio cues through bone conduction will eliminate reliance on hard-to-hear audio cues from maps creating a calmer and safer riding experience. The improved perception of safety will encourage more frequent and longer rides, especially in urban settings.

2. Multifunctional Glasses

All-weather usability enables cyclists to ride comfortably in rain, sun, wind, and cold without performance loss. Integration of navigation, wind protection and sun protection into one device delivers greater physical comfort and reduced gear management. Reduced need for multiple devices improves spontaneity and ease of use for daily commuting and leisure rides. Additionally, a lightweight and ergonomic design will enhance comfort and reliability during long-distance riding.

3. Affordable, Simple Smart Glasses

A lower purchasing cost makes smart navigation technology accessible to everyday cyclists, not just tech enthusiasts. A simple setup and an intuitive interface improves user confidence and reduces technological anxiety. Focusing on essential cycling functions ensures higher reliability and easier adoption. Users benefit from strong cost–value satisfaction without sacrificing safety or performance features.

Brand Identity

RideAware is a smart-mobility brand that empowers cyclists and micro mobility users with intuitive, distraction-free navigation through augmented reality and audio prompts. Our AR glasses project subtle turn-by-turn cues directly into the rider's immediate line of sight, complimented by bone-conduction audio prompts that keep ears open which enhances awareness. By combining clarity, safety, and seamless everyday usability, RideAware transforms ordinary rides into confident, fluid journeys.

Our brand blends functional innovation with an approachable, modern aesthetic that speaks to a growing community of riders who value safety, efficiency, and active, sustainable urban living. Our design language focuses on light, precise, and trustworthy technology that feels supportive rather than intrusive. With photochromic, wind protection and prescription-ready lenses, RideAware integrates effortlessly into the rider's routine, replacing multiple accessories with one smarter alternative.

> "If navigation stayed right in my line of sight and worked in any weather, I'd feel much more confident."
>
> - **Erik Gellert**

At RideAware, we stand for smarter streets, empowered riders, and a culture in which navigation enhances, rather than distracts from, the riding experience. We support safety-minded commuters, performance-driven cyclists, and health-conscious riders who want to optimise every journey to see beyond.

Product and Service Feasibility

Customer Definition

Understanding RideAware's early customers is crucial for assessing product-market fit. Interviews and cyclist behaviour research point to three key early-adopter segments, each with distinct needs, motivations and willingness to pay.

Segment 1: Safety-Driven Urban Commuters

Profile: Daily commuters and high-frequency urban riders such as food delivery couriers, who operate in dense traffic, navigate constantly changing routes and ride under time pressure.
What they value: They seek greater situational awareness, distraction-free navigation and a calmer, more confident riding experience supported by reliable, weather-resistant equipment.
Purchase drivers: Their main motivation is risk reduction; fewer near misses, lower mental load and more predictable traffic flow. For delivery riders, staying focused and navigating frequent route changes safely is especially critical.
Reasons for hesitancy: Some hesitate due to concerns about technical complexity, reduced visibility in rain or glare, and doubts about durability or battery life.
Price sensitivity: Their willingness to pay varies (€120–€180) , though employers or platforms may subsidise safety equipment.

Segment 2: Performance-Oriented Recreational Riders

Profile: Road cyclists, triathletes and long-distance riders who train frequently and rely on navigation during high-speed or extended rides.
What they value: They prioritise precise, low-latency navigation, reliable hazard detection on rural or unfamiliar routes, and lightweight, aerodynamic equipment that does not disrupt their riding flow. They value multifunctionality, so the glasses should replace their usual sunglasses.
Purchase drivers: Their main motivation is performance optimisation. They value technology that helps them maintain forward focus, manage complex routes efficiently and avoid unexpected obstacles without losing speed.
Reasons for hesitancy: They expect high technical accuracy and would expect additional performance metrics and worry about battery life.
Price sensitivity: They are willing to pay €180–€250 for clear performance benefits.

Segment 3: Tech-Savvy Urban Mobility Users

Profile: Younger riders, e-bike commuters and multimodal urban travellers who use cycling as an integrated part of their daily mobility and value modern design and digital connectivity.
What they value: They seek multifunctional, minimalist and aesthetically appealing eyewear that combines navigation, alerts, sun protection, and integrates seamlessly with smartphone apps.
Purchase drivers: Their adoption is driven by convenience and identity alignment. They want intuitive, modern technology that enhances their daily routine and fits their lifestyle.
Reasons for hesitancy: They may hesitate due to concerns about frequent charging, potential over-complexity and uncertainty about long-term durability.
Price sensitivity: They have a moderate to high willingness to pay (€120–€200), depending on perceived utility and design quality.

Among these, safety-driven urban commuters represent the primary target for early market entry due to high problem awareness and consistent daily usage.

Overall willingness to pay across segments ranges from €120 to €250, depending on perceived performance and safety benefits.

Casual leisure riders and low-budget cyclists were excluded as core targets, as their navigation and safety needs are less frequent and their willingness to pay is significantly lower.

These customer insights directly inform RideAware's design priorities, such as minimal visual distraction, lightweight hardware, reliable hazard detection and seamless app integration.

Expert Opinion on Technical Feasibility

As mentioned, to assess the technical feasibility of our product, we conducted surveys with three experts specialized in human-computer interaction, embedded systems, computing, and AR/VR. The purpose of this primary research was to identify flaws in our current concept model and highlight key risks we needed to address.

The experts' opinions varied, but there was a consensus that the current concept had significant tech and engineering burdens. Integrating rear sensors, VIO SLAM, and an always-on AR HUD within a lightweight frame, while technically feasible, would significantly increase unit cost and make the product commercially infeasible. Dr. Molloy noted this was mainly due to high computing needs requiring a more expensive Nvidia edge module. There were also concerns about reliability, as the sensors in our original model are not sufficiently capable of detecting objects in busy environments.

Decision to Pivot Our Concept

A key theme was user trust and legal liability from positioning RideAware as a safety device. The experts strongly advised narrowing our scope and focusing on a high-quality user experience around a small number of technically realistic functions—specifically navigation delivered through audio and AR. Based on this, we believe that pivoting to a navigation-focused product is the best way to proceed for our MVP. (Please see appendix A for further details).

Product Design of Pivoted Model

This new pivot in our core technology means slight changes to the product design and service. Firstly, the new navigation-focused AR frames will use TR90 clip-on photochromic lenses to keep weight low, maintain a high level of durability, and enhance visibility under high UV environments.

Display

For the AR component, a waveguide-based optical see-through head mounted display (OST-HMD) will project simplified turn-by-turn cues into the rider's upper field of vision to ensure the rider is not distracted while cycling.

Research in the space of AR and HMDs state that legibility in bright conditions continue to be an issue (Qian et al., 2024). To address this, we plan to use a micro-display that utilizes LCoS (Liquid-crystal-on-silicon) technology for a bright display, in addition to ambient sensing technologies for adaptive brightness that will enhance display transparency outdoors (Kim et al., 2022; Erickson et al., 2020).

Navigation and Processing

Processing of navigation will run on the smartphone, and the integrated MCU with BLE will render the cues that it receives from the phone as AR overlays.

Audio

For the audio cues, open-ear speakers that use bone-conduction technology will send audio prompts to the user at a capped volume to maximise situational awareness while cycling (Chang-Geun et al., 2011).

Electronics Integration

Lastly, the battery and electronic components will all be placed along the temples of the frames to ensure weight is distributed evenly, aligning with work suggesting that weight of the glasses can strongly affect comfort and fatigue (Ito et al., 2021)

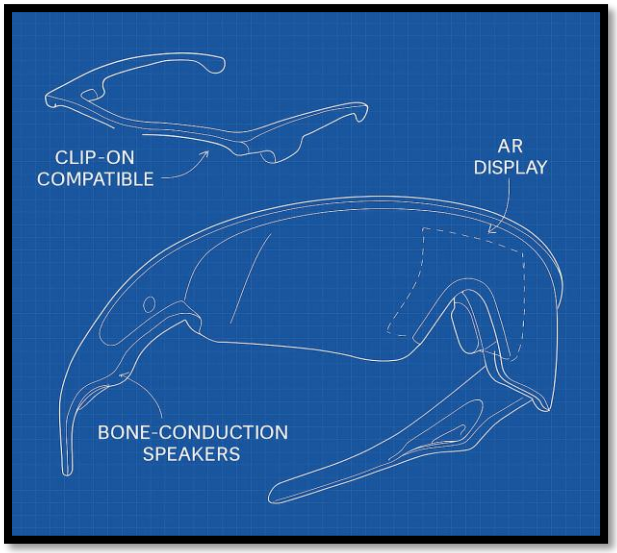


Figure 2: MVP prototype based on consumer feedback.

Limitations of Pivoted Model

The *pivoted model* is much more feasible, but some concerns remain:

Display: Existing AR waveguide combiners still struggle with contrast management and colour washout in transparent views, creating a risk that riders miss cues in bright conditions (Kim et al., 2022; Erickson et al., 2020).

Latency: BLE latency and jitter can cause navigation indicators to appear too late, so cues and computation both need to be as simple and robust as possible (Tosi et al., 2017).

Power and Thermal: Combining high-brightness displays, continuous BLE, and audio in a lightweight frame raises power and thermal constraints that must be managed to avoid overheating and discomfort over longer rides (Kim et al., 2022).

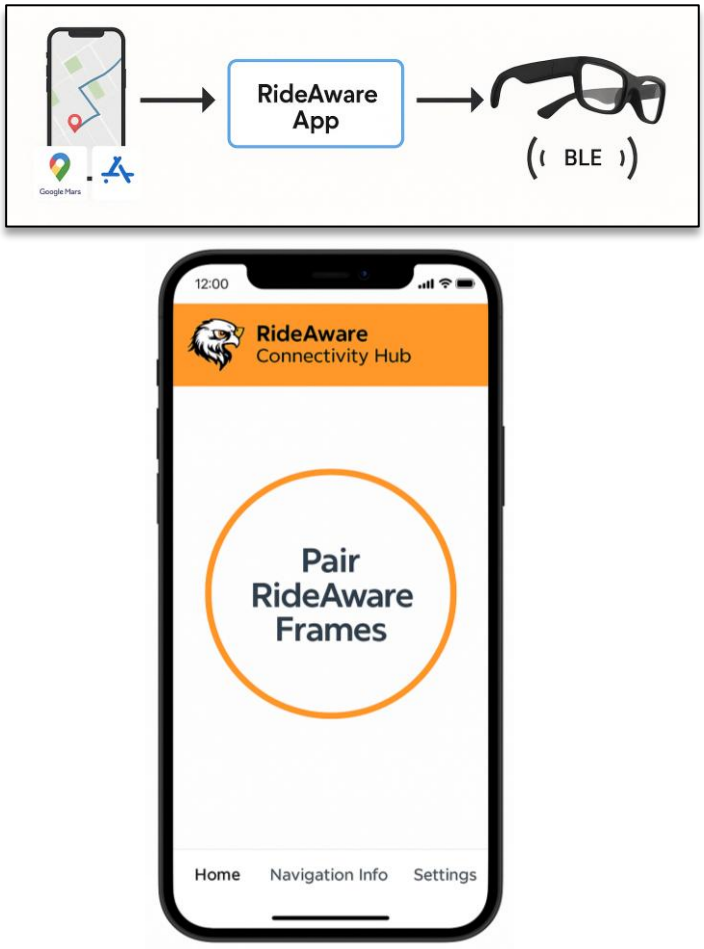


Figure 3: Google and Apple Mapping APIs.

Google and Apple API Integration

With the pivot away from the SLAM technology and rear sensing safety, a big component of our value proposition centers around navigation. To ensure that we can provide our customers with a quality navigation experience, we intend to integrate Google and Apple Map APIs into our technology proposition.

How This Looks In Practice

Users will need to use navigation APIs such as google maps or apple maps within their smartphones to communicate real-time directions to the frames. To use these APIs, RideAware must comply with the terms and conditions for both Google and Apple. For both platforms, it is permissible to use their APIs at a free capacity with the use of an app that is compliant with their terms and conditions (Google Maps Platform Pricing, 2025; Apple Mapkit JS Documentation). However, both APIs will charge licensing fees if the usage climbs beyond their free usage caps which differ for both platforms (Google Maps Platform, 2024; Apple Developer Program 2025).

However, during the current start-up stage and the development of our MVP, we do not foresee exceeding these usage caps. In the future, if usage were to exceed these caps, we would then factor that into the COGS, and consider a monthly subscription to gain access to the app.

Under this model, we will develop the RideAware App that will draw on Google/Apple mapping API's which will grant us access to critical navigation data. From there, our proprietary application will be used to compute the map route, turn by turn directions, and then send derived cues to the RideAware frames via BLE (Afi Labs, 2025). This improves feasibility as the physical product will not be open to API usage or licensing implications directly, and the usage caps will only be applied to the RideAware app.

Product and Service Feasibility

Manufacturing Feasibility

Manufacturing of the pivoted navigation-focused frames is feasible as all the major subsystems we're proposing to implement already exist in other smart-glasses products:

Alova Audio | Shenzhen
A smart glasses manufacturer in Shenzhen, for example, already manufacturers glasses that have open-ear directional speakers, BLE tech, magnetic charging, and a slim form-factor that is IP67 rated (Alova Audio, 2025).

Gonbes Technology | Shenzhen
Also manufactures smart glasses with the same technology at mass scale, highlighting the existence of mature supply chains for glasses with audio systems (Gonbes, 2025).

Shenzhen Anpo Intelligence Tech Co. | Shenzhen
In terms of OEM/ODMs that are more experienced in AR integration, Shenzen Anpo Intelligence Tech Co. is an established manufacturer that has experience developing AR micro displays that use LCoS (Enmesi, 2025).

Strategic Manufacturing Partner: Goertek

To successfully get our product to market, we identified Goertek as the best long-term manufacturing partner for the RideAware frames. Goertek is an established manufacturer in the production of wearable technology and are behind the flagship Ray-Ban Meta smart glasses (Financial Times, 2025). The company has experience with both VR/MR head mounted displays and AR/AI glasses. Production services for their clients cover industrial design, optics, acoustics, computing and software, moulding, and assembly & testing (Goertek, 2025). Their development and ODM knowledge is exactly what RideAware requires to develop the AR navigation frames.

Goertek's proven track record in the wearable technology industry, along with their deep knowledge on waveguide displays, acoustic integration, and mass volume manufacturing makes them the strongest strategic partner for RideAware. In the short term, however, we do foresee the possibility of partnering with smaller OEMs like Shenzen Anpo Intellence Tech to develop an MVP or conduct early-stage prototyping.

Table 1: Projected RideAware Components Needed for Frames (Display Module, 2023).

Component	Function
Grilamid TR 90 Frames with clip-on compatibility	Will act as the light-weight frame for our wearable technology, while allowing for prescription lenses to be attached with integrated clip-on magnets. The frame will also be water resistant up to a rating of IP67
Photochromic Lenses	UV-dependent tint will allow maximal comfort on the eye based on the changing outdoor conditions.
Waveguide Projector	Will project navigation cues from a micro-display and will use a transparent AR overlay to ensure minimal distraction.
MCU with BLE	This is the computing unit that will be used to communicate with the smartphone via Bluetooth. Will also serve as the processor that interprets navigation data, coordinates display, sensing, and audio.
Bone Conduction Speakers	This will act as the open ear sound system. It will allow cyclists to receive auditory navigation prompts through temple vibrations, maximising awareness of surrounding traffic
IMU Sensors	Will act as a stabilising mechanism for the AR display as the sensors will track head motion and orientation. This will also facilitate the process of ensuring world-locked AR.
Lithium-ION Battery	Will supply power to the RideAware frames, providing several hours of usage per charge.
Flexible Printed Circuit Board (PCB)	This is where the entire system will live. The PCB will connect all the electrical components within a flexible board that can be wrapped routed throughout the frames.

Distribution Strategy

Due to the high-tech nature of the product, we believe that maintaining a physical presence is very important for acquiring consumer trust and lessening adoption friction. For this reason, RideAware will proceed with a dual distribution method under a direct-to-consumer model that utilizes e-commerce channels and specialist retail bike shops.

Company Website Powered by Shopify

For the online component, we will use a branded Shopify store as our primary distribution channel. Opting for this will enable us to build brand awareness and facilitate the establishment of RideAware through a solidified digital presence. This is a strong starting point as it will enhance control over pricing, access to customer insights & analytics, and will facilitate the ability to market the glasses through social channels (Shopify, 2025). This will also act as the central platform that consumers can use to log customer service concerns or get an in-depth overview of the product, which will support long-term marketing efforts.

Supplementary Distribution with Amazon FBA

Amazon FBA will act as the secondary channel for our online distribution. We intend to use this as a supplementary channel for a smaller batch of SKUs. The rationale for this is that while FBA provides access to Prime customers, rapid shipping, and an extensive logistics infrastructure, it also reduces scaling flexibility and means higher storage fees (Griffin, 2025). However, by readily having a smaller batch of frames available through FBA, we can gauge demand of the product, improve product visibility, and benefit from an efficient distribution process as the firm grows its user base.

Vendor:	Amazon	Shopify
Fulfilment Per Unit	$3 (Assuming Standard)	Variable Usage Based ($29/month)

Table 2: Sample Fulfilment Pricing ((Wells, 2025; Harvey, 2025).

Retail Channels

For the offline and retail component of our distribution strategy, we intend to partner with speciality bike shops across Dublin. We will first commence with a few independent bike shops that will provide shelf space for the RideAware frames in exchange for a 10-15% share of our profit margin. In the short run, this means a profitability trade off, but building partnerships with reputable bike shops across the city will enhance our credibility and consumer trust. Additionally, these retail partners will act as certified dealers that can assist with after-sales support.

Figure 4: Visual Representation of our Projected Distribution Route (CHN – IE).

Implementation of Distribution Strategy

In practice, this dual distribution strategy will involve a few steps. The first is shipping from the factory in China to an EU-based 3PL that will hold RideAware inventory in bulk. From here, the third-party logistics provider will ship Shopify orders directly to customers; ship wholesale orders to bike shops; and periodically ship batches to Amazon fulfillment centres based on demand (Amazon MCF, 2025). This strategy will enable RideAware to reach customers through both digital and in-person channels.

Industry Attractiveness

Market Overview and Growth

AR & Smart Wearables

The global wearables market is growing rapidly and is expected to increase by USD 12B between 2024 and 2029, reaching USD 50B by 2029 (Euromonitor International, 2025). Long-term forecasts predict a compound annual growth rate (CAGR) of 19.5% up to 2035, driven by rising demand for integrated smart wearables (Future Market Insights, 2025). Smart wearables are expanding into new form factors, such as smart glasses. As AR becomes more mainstream, global AR eyewear shipments are projected to reach 14.12 million units by 2025 (Euromonitor, 2025; IDC, 2025). Products such as the Meta Ray-Ban and the Vuzix Blade demonstrate increasing consumer acceptance of AR-enabled eyewear.

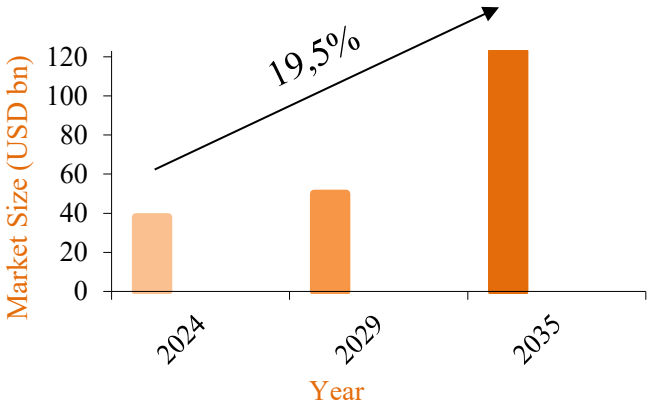


Figure 5: Global Wearable Market Growth Forecast.

Smart Cycling Technology Market

The global bicycle safety technology market is expected to grow from USD 2.47B in 2024 to USD 6.05B by 2033 (CAGR 10.2%) (Market Research Future, 2024). This growth is driven by urbanisation, rising health awareness and pro-cycling policies. Increasing adoption of IoT devices, connected wearables and smart helmets strengthens demand for hands-free, real-time navigation, creating favourable conditions for AR solutions like RideAware. Innovation in connected cycling tech, such as Garmin's solar-powered, multi-band GNSS Edge 1040 illustrates rising user expectations (Fact.MR, 2025). European subsidies and infrastructure investments further boost demand for premium navigation and AR wearables.

Navigation & Safety Market

The navigation and mapping devices market, dominated by bike computers from brands such as Garmin, Wahoo and Hammerhead, as well as smartphone-based systems, continues to grow. Safety remains a major behavioural driver, with the increased risk of accidents in low visibility conditions driving demand for smart lights, visibility-enhancing helmets and other protective accessories (Fact.MR, 2025).

Target Market

Total Addressable Market (TAM)

The TAM includes all cyclists and micro-mobility users who rely on digital navigation. Europe is a particularly attractive market, with an estimated 11.7 million bicycles in circulation and 5.1 million e-bikes in use across the EU (CONEBI, 2025). Bicycle production reached 14.7 million units in 2022, marking a 29% increase since 2012 (Fact.MR, 2025). Most cyclists use smartphone apps or GPS devices to navigate, creating a substantial TAM for hardware that enhances navigation, such as AR cycling glasses (Cycling UK, 2025).

Serviceable Available Market (SAM)

The SAM focuses on European cyclists who actively use digital navigation tools and show a willingness to adopt premium technology to improve comfort and usability. This includes urban commuters in major cycling cities (e.g. Dublin, Berlin, Amsterdam), road cyclists and triathletes, recreational fitness riders, urban food delivery services, and the fast-growing e-bike community (CONEBI, 2023). These tech-affine, premium-oriented subgroups represent the portion of the market that RideAware can realistically serve.

Serviceable Obtainable Market (SOM)

The SOM reflects RideAware's early-market reach among tech-affine cyclists, early adopters in strong cycling cities, Garmin/Wahoo users and urban commuters. These groups match early-adopter profiles described by innovation diffusion theory and emerging hardware adoption patterns (McKinsey & Company, 2023; Rogers, 2003). Through pilots, clubs and cycling events, RideAware can target a small but commercially meaningful early-adopter segment in European markets.

Competitive Landscape in Europe

The European smart cycling technology market can be grouped into four segments: Market Leaders, Runner-Up Brands, Medium-Sized Players, and Minor Players. These categories reflect the differences in market maturity and technological capability within the emerging AR eyewear segment.

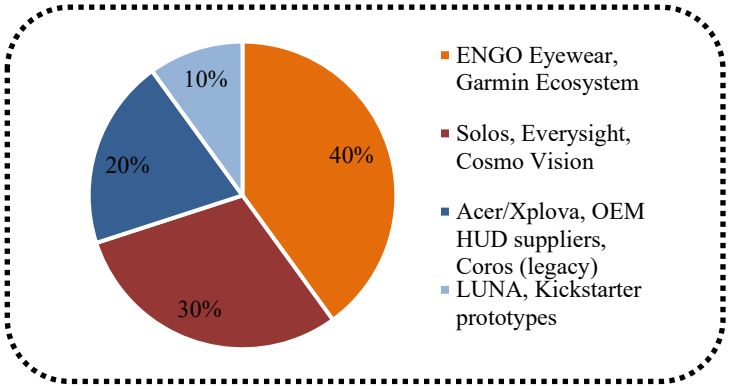


Figure 6: Estimated Competitive Segments in the AR Cycling Glasses Market.

Market Leader

ENGO is Europe's leading specialist in AR-enhanced cycling glasses, widely adopted by road cyclists and triathletes. Its leadership position is solidified by its heads-up display technology, ANT+ and Garmin connectivity, and strong partnerships in performance cycling. Although Garmin no longer offers the Varia Vision headset, its broader ecosystem, which includes Edge computers, Varia radar, smart lighting and sensors, dominates the cycling technology market. Garmin sets the benchmark for integration, data reliability, and user expectations, making it a powerful indirect competitor for navigation-focused AR products.

Runner-Up Brands

Runner-up brands include Solos, Everysight Raptor and Cosmo Vision. Solos offers lightweight HUD displays and voice assistance but has limited market penetration in Europe, serving mainly as a usability benchmark. Everysight's discontinued Raptor remains a key technological reference due to its advanced AR overlays. Cosmo Vision targets urban cyclists with hybrid AR features focused on navigation and visibility, making it one of the closest functional comparisons for RideAware.

Medium-Sized Players

Medium-sized players include Xplova/Acer, OptoTech and other OEM HUD manufacturers, as well as earlier smart glasses models from Coros. Xplova/Acer has developed cycling technology and experimented with AR prototypes; while their presence is stronger in Asia, they could enter the European AR segment if strategically aligned. OptoTech and various OEM suppliers shape the market indirectly by providing AR modules and HUD components, lowering entry barriers for new entrants. Coros, although no longer active in AR eyewear, helped shape early expectations around design, audio integration and battery performance.

Minor Players

Minor players include LUNA Smart Cycling Glasses, various Kickstarter and Indiegogo HUD concepts, and low-cost OEM smart glasses from Shenzhen. LUNA is a crowdfunded prototype with basic illumination and limited AR capability, low production volume and no ecosystem integration, making it commercially insignificant but indicative of low-end experimentation. Crowdfunded HUD concepts frequently introduce novel ideas but lack stable production, durability and market relevance. Similarly, low-cost OEM smart glasses offer basic AR functionality and compete purely on price, positioning them far from RideAware's premium, reliability-focused segment.

Conclusion

The competitive landscape is fragmented, with ENGO and Garmin shaping high-performance standards while other players test early AR concepts. Yet, there is a clear gap for simple, navigation first AR glasses, tailored for everyday, commuter and road cyclists. By focussing on intuitive, distraction-free navigation, RideAware can meet this need and establish a clear market position in the evolving cycling technology market.

Market Entry and Future Market Expansion

Recent survey data shows that more than half of adults in Ireland say safety concerns prevent them from cycling. Common barriers include heavy traffic, a lack of segregated bike lanes, and poor infrastructure (RedClick, 2025). However, through our conducted interviews, we could identify growing interest in biking as an exercise and a sustainable mode of transport. Additionally, the Irish government has shown strong institutional support through investments since 2020. This signals favourable public policy conditions for active travel and potential for partnerships (National Transport, 2024). RideAware operates within the emerging AR/XR wearable devices market, which represents the next generation of wearables beyond smartwatches and sports trackers. RideAware's AR glasses fall under the category of head-mounted, high-power wearables, delivering heads-up, hands-free information through turn-by-turn navigation displayed directly in the cyclist's field of vision.

Challenges

Entering this market will present several challenges that need to be addressed when launching. Wearable hardware, especially premium eyewear with AR, typically requires substantial marketing and customer-acquisition expenditures to overcome consumer hesitancy. Hardware startups often face high consumer acquisition costs, which must be carefully budgeted into unit economics. As a head-worn AR device for cyclists interacting in traffic, RideAware must address safety, reliability, and liability concerns. Rigorous testing, clear usage guidelines and possibly regulatory approval or certification will be necessary. The product will compete not only with conventional cycling eyewear and helmets, but also with emerging smart glasses / AR wearables. To succeed, RideAware must clearly demonstrate the unique value of simple AR-enhanced navigation and safety features for cycling, rather than relying on novelty alone. To reach cyclists, RideAware will likely need a mix of distribution channels: direct-to-consumer online sales, speciality bike shops, sports/outdoor retailers, and possibly collaborations with local authorities or cycling organisations.

These risks underline the need for conservative financial planning, pilot programmes, and clear communication of safety and value to customers.

Following initial market entry in Ireland, RideAware could pursue expansion into the DACH region (Germany, Austria, Switzerland). Germany represents one of Europe's largest cycling and e-bike markets, supported by highly developed cycling infrastructure, strong environmental awareness, and sustained demand for premium cycling equipment (ZIV, 2025). Expansion into the UK represents a further logical step, driven by similar trends in micromobility growth and urban cycling safety initiatives. These regions combine strong purchasing power with mature or rapidly growing cycling cultures, which aligns well with RideAware's premium positioning.

Future Product Range Expansion

RideAware plans to address varying aesthetic and functional preferences through multiple design variants and potential collaborations with established eyewear manufacturers. Application integration with platforms such as Komoot and Strava will further strengthen its ecosystem. Additionally, a performance-oriented "Pro" model with advanced tracking, analytics, and training features would target competitive and sport-focused cyclists.

Our Positioning

RideAware offers a unique combination of AR-enabled navigation, reducing the need for mircomobility users to glance down at their phones or bike computers, increasing safety and situational awareness. RideAware leads competitors in ease of use, navigation performance and safety (see Appendix 2). The premium sports-eyewear / cycling-eyewear design, addresses both functional and aesthetic demands of cyclists who value performance, comfort, and style. They appeal to sustainability-conscious and health-oriented consumers, combining efficient transportation with safety and convenience for everyday commutes or leisure rides. This value proposition allows RideAware to target commuters, urban cyclists concerned about safety, and sporty/outdoor-oriented users seeking premium eyewear. By combining AR-based functionality with established cycling eyewear channels, RideAware can differentiate itself in a crowded cycling tech market and tap into broader wearable tech trends

Organisational Feasibility

Overview of Team Skills

In order for our product to be successful, a variety of skills are required. To determine what skill sets our team possesses, we collectively completed a survey. Based on the sectors listed below, each team member rated themselves on a scale of one to five, with one being little to no skills in that area and five feeling competent and skilled in that field.

Business Skills	Interpersonal Skills
Financial Literacy	Leadership
Marketing Skills	Problem Solving
Operations Knowledge	Time Management
Technological Proficiency	
Product Development	
Legal Knowledge	

Table 3: Business and Interpersonal Skills Table.

Outline of Planned Positions

Currently, RideAware is planned to have the following *seven positions*:

- ***Chief Executive Officer CEO*** - Responsible for overall company strategy
- ***Chief Operating Officer COO*** - Focused on supplier relations and supply chain management
- ***Chief Financial Officer CFO*** - Overseeing company budgeting and financing operations
- ***Chief Marketing Officer CMO*** - Leads advertising and company branding
- ***Chief Technical Officer CTO*** - Oversees future product development
- ***Lead Software Developer*** - In charge of back-end and front-end coding and software design
- ***Administrative Assistant*** – In charge of administrative tasks and customer support

Existing Skill Gaps

After performing the survey, our group identified three skill gaps in product development, technological proficiency and legal knowledge. The maximum possible score for each skill is 25, and the group's average skill level is 16. In technological proficiency, the group scored 11/25, while product development has the smallest skill gap, with our team scoring 13/25. The largest skill gap lies in legal knowledge, where our group scored 8/25.

Name	Financial Literacy	Marketing Skills	Operations	Tech Proficiency	Product Development	Legal Knowledge	Leadership	Problem Solving	Time Management	Average Skill Level
Oliver CTO	3.00	2.00	4.00	4.00	3.00	1.00	4.00	4.00	3.00	**3.00**
Aidan COO	4.00	3.00	3.00	3.00	2.00	2.00	3.00	4.00	3.00	**3.00**
Madeline CMO	2.00	5.00	5.00	1.00	3.00	1.00	3.00	4.00	4.00	**3.11**
Juliana CEO	3.00	4.00	5.00	1.00	3.00	1.00	4.00	4.00	4.00	**3.22**
Daniel CFO	5.00	3.00	3.00	2.00	3.00	3.00	4.00	5.00	5.00	**3.67**
Skill Total	**17.00**	**17.00**	**20.00**	**11.00**	**13.00**	**8.00**	**18.00**	**21.00**	**19.00**	**16.00**

Skill Gap

Table 4: Skills Gap Table.

Organisational Feasibility

Organisational Structure

RideAware will incorporate as a private limited company (LTD) within Ireland. This structure ensures that for a small company like a start-up, with increased risk of failure, any liability is limited by investment in the organisation. (Irish Formations, n.d.).

Ride Aware will seek to utilise a flat reporting structure to ensure that communication channels remain open (Carzo, 1969). Flat reporting structures allow for improved communication and reduce levels of hierarchy within an organisation. For a new start-up, this is crucial. Ride Aware will have seven employees, the five executive positions, in addition to a lead software developer and an administrative assistant. Given the small team, unnecessary levels of hierarchy would only slow communication and burden the organisation.

Addressing Skill Gaps

Since the skill gap in product development is small, our team intends to focus inwards and improve our product development skills. Through initial product planning and research, our team has already gained more knowledge on how the product can be developed. Additionally, Ride Aware currently plans to partner with manufacturing company Goertek. As described in the product feasibility section, Goertek is a leading smart glasses manufacturer based in China. Partnering with Goertek allows RideAware to acquire a knowledgeable and experienced partner who can ensure that the manufactured glasses will be of high quality. Additionally, Goertek can ensure that potential oversights in the product can be fixed before mass production begins.

One of the largest challenges in the development of the product is an application and interface that allows the glasses to interact and sync with the user's mobile phone. This is a software and coding-intensive process that our team currently lacks competency in as shown by the skill gap in technological proficiency. Commissioning an app holds immense costs that are currently unrealistic for our business. To address this skill gap, we recruited additional talent.

We focused on younger professionals who would be willing to accept company stock in exchange for their work. We identified a friend of a group member named Nabil Hussein. Nabil has a master's degree in computer science and a bachelor's of science from Griffith College. Nabil agreed to the terms of back-end and front-end design as our lead software developer for a 10% stake in the organisation. Nabil offers a realistic way to fill our group's skill gap while ensuring excess funds are conserved for other purposes.

Navigating patent law for both, the patenting of our product, and ensuring our product avoids patent infringement on other smart glasses would require legal counsel who is versed in patent law. Murgitroyd is a multinational company specializing in patent law. Operating across Europe, the Americas and Asia. With an office located in Dublin, Murgitroyd's services can ensure that RideAware's patents are filed and don't infringe on existing smart glasses patents. Murgitroyd also operates in trademark law and can ensure that Ride Aware's name and logo are trademarked. Murgitroyd offers advantages as a multinational company that can process trademarks and patents across all of Europe as RideAware looks to expand. (Murgitroyd, 2025).

Customer Service Strategy

RideAware's customer service strategy is based around digital contact channels such as the company email and live chat on the website. These digital channels will be monitored by RideAware's administrative assistant.
In instances where it is deemed that a product has to be replaced or returned, arrangements will be made on a case-by-case basis to ensure customer satisfaction.

Since RideAware products will be sold through physical bike stores, returns would often have to be done through the third-party retailers where the product was purchased, based on their return policies.

Financial Feasibility

Assumptions

Given the complexity and nature of our product and business, several assumptions were required to conduct a relevant and useful financial analysis. Firstly, we have assumed a just-in-time (JIT) inventory system for both efficiency and simplicity purposes. This assumption indicates that the number of units produced approximately equals the number of units sold, which aided in simplifying our process overall. Specifically, our statement of cash flows and cost of goods sold through a more direct cost flow. In addition, as a partial strategic business decision, we have opted for a fully remote corporate team to minimize costs in the early startup stages. This implies significant savings on office rent and other expenses associated with an in-office work environment. Next, we have eliminated credit sales to customers and credit purchases from suppliers (i.e., cash sales and payments only), strictly to reduce the complexity associated with variability in supplier and customer credit terms.

In terms of our monthly sales for year one of operations, our team used our original estimates explained in our concept paper and applied a positive linear growth pattern from month 1 until month twelve (see *Appendix 3.3* for further details). Also, we assume double taxation holds between Ireland and China in the case of RideAware's operations (Office of the Revenue Commissioners, 2000). However, it is also useful to note that a full exemption may be possible given RideAware's startup nature (Localenterprise.ie, 2024). For depreciation and amortisation assumptions please see *Appendix 3.1, 3.2*.

Financial Statements

Profit and Loss Statement (P&L)

Despite the typical loss-making nature that is heavily common amongst startups, we primarily expect to be operating at a profit (i.e., in months 2-12). This can be attributed to intentional cost-saving measures, such as our remote corporate work approach.

RideAware Ltd.
Sample Income Statement
For the Year Ended 31 December 2027

Line Item	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9	Month 10	Month 11	Month 12	Total
Net Sales	€ 21,728.00	€ 23,668.00	€ 25,608.00	€ 27,548.00	€ 29,488.00	€ 31,428.00	€ 33,368.00	€ 35,308.00	€ 37,248.00	€ 39,188.00	€ 41,128.00	€ 42,292.00	€ 388,000.00
Cost of Goods Sold (Includes Labor and Shipping)	€ 16,688.00	€ 18,178.00	€ 19,668.00	€ 21,158.00	€ 22,648.00	€ 24,138.00	€ 25,628.00	€ 27,118.00	€ 28,608.00	€ 30,098.00	€ 31,588.00	€ 32,482.00	€ 298,000.00
Gross Profit	€ 5,040.00	€ 5,490.00	€ 5,940.00	€ 6,390.00	€ 6,840.00	€ 7,290.00	€ 7,740.00	€ 8,190.00	€ 8,640.00	€ 9,090.00	€ 9,540.00	€ 9,810.00	€ 90,000.00
Operating Expenses													
Production Warehouse Rent (Average monthly cost @ RMB 33.8 or €4.12 x 400 sq m)	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 1,648.00	€ 19,776.00
Insurance Costs (€2000 yearly- see concept paper)	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 166.67	€ 2,000.00
Utilities Expenses (Assume @ 10% of total warehouse cost)	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 164.80	€ 1,977.60
Marketing Costs (assumed in Concept paper)	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 500.00	€ 6,000.00
Product Branding Expense (assumed in concept paper)	€ 4,300.00	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ 4,300.00
Depreciation Expenses													
Manufacturing Equipment	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 14,000.00
Computers/Software (e.g., Macbooks)	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 2,300.00
Employee Phones	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 1,030.00
Injection Mould	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 880.00
Amortization Expenses													€ -
Prototypes	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 14,000.00
Website	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 700.00
Patents/Trademarks	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 790.00
Patent Renewal Expense (€300/year)	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ 300.00	€ 300.00
Total Operating Expenses	€ 9,587.80	€ 5,287.80	€ 5,287.80	€ 5,287.80	€ 5,287.80	€ 5,287.80	€ 5,287.80	€ 5,287.80	€ 5,287.80	€ 5,287.80	€ 5,287.80	€ 5,587.80	€ 68,053.60
Operating Income (EBIT)	-€ 4,547.80	€ 202.20	€ 652.20	€ 1,102.20	€ 1,552.20	€ 2,002.20	€ 2,452.20	€ 2,902.20	€ 3,352.20	€ 3,802.20	€ 4,252.20	€ 4,222.20	€ 21,946.40
Interest Expense	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	
Income before taxes (PBT)	-€ 4,547.80	€ 202.20	€ 652.20	€ 1,102.20	€ 1,552.20	€ 2,002.20	€ 2,452.20	€ 2,902.20	€ 3,352.20	€ 3,802.20	€ 4,252.20	€ 4,222.20	€ 21,946.40
Income Tax Expense (IE)	€ -	€ 25.28	€ 81.53	€ 137.78	€ 194.03	€ 250.28	€ 306.53	€ 362.78	€ 419.03	€ 475.28	€ 531.53	€ 527.78	€ 3,311.78
Income Tax Expense (CHN)	€ -	€ 50.55	€ 163.05	€ 275.55	€ 388.05	€ 500.55	€ 613.05	€ 725.55	€ 838.05	€ 950.55	€ 1,063.05	€ 1,055.55	€ 6,623.55
Net Income	-€ 4,547.80	€ 126.38	€ 407.63	€ 688.88	€ 970.13	€ 1,251.38	€ 1,532.63	€ 1,813.88	€ 2,095.13	€ 2,376.38	€ 2,657.63	€ 2,638.88	€ 12,011.08
Retained Earnings	-€ 4,547.80	-€ 4,421.43	-€ 4,013.80	-€ 3,324.93	-€ 2,354.80	-€ 1,103.43	€ 429.20	€ 2,243.08	€ 4,338.20	€ 6,714.58	€ 9,372.20	€ 12,011.08	€ 12,011.08

Table 5: Sample Profit and Loss Statement FY2027.
Note: *As explained previously in our concept paper, our approximate unit cost is €149*

Financial Feasibility

Also, our team has opted to completely reinvest any profits earned back into the business instead of taking consistent wage drawings. Therefore, line items such as corporate salaries and payroll taxes typically seen in the operating expenses section of the profit and loss statement are negligible. Therefore, at the end of our first fiscal year (FY2027), we expect an EBIT of €21,946.40 and net income of €12,011.08.

Statement of Cash Flows

As explained by HSBC (2025), "a cash flow statement tracks how much money your business brought in over a set period, and how much was spent." In our case, we expect to be net cash flow negative from months 1 through 9 due to heavy investing activities being assumed in month 1.

This includes essential operational purchases such as manufacturing equipment, company phone purchases, prototypes, patents, web development, and injection moulds. However, these outflows from initial investment activities will be partially offset by the depreciation and amortisation add-backs. In addition, from the end of month 9 to month 12 we expect to be net positive cash flow overall, as net cash from operations starts to increase from heightened product interest (see *Appendix 3.3* for further details and visuals) and significant cash disbursements start to be covered.

Looking forward, we expect cash flows to remain positive in upcoming fiscal years, contingent on factors such as continuous stable sales growth and product interest, as well as reasonable operating expenses and capital expenditures.

RideAware Ltd.
Sample Statement of Cash Flows
For the Year Ended 31 December 2027

	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9	Month 10	Month 11	Month 12	Total
Cash Flow from Operations													
Net Income	-€ 4,547.80	€ 126.38	€ 407.63	€ 688.88	€ 970.13	€ 1,251.38	€ 1,532.63	€ 1,813.88	€ 2,095.13	€ 2,376.38	€ 2,657.63	€ 2,638.88	€ 12,011.08
Additions to Cash													
Depreciations/Amortizations													
Manufacturing Equipment	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 14,000.00
Computers/Software (e.g., Macbooks)	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 191.67	€ 2,300.00
Employee Phones	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 85.83	€ 1,030.00
Prototypes	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 1,166.67	€ 14,000.00
Website	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 58.33	€ 700.00
Patents/Trademarks	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 65.83	€ 790.00
Injection Mould	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 73.33	€ 880.00
Decrease in A/R	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -
Increase in A/P	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -
Increase in taxes payable	€ -	€ 75.83	€ 244.58	€ 413.33	€ 582.08	€ 750.83	€ 919.58	€ 1,088.33	€ 1,257.08	€ 1,425.83	€ 1,594.58	€ 1,583.33	€ 9,935.33
Subtractions from Cash													
Increase in inventory (Assume a JIT system for optimization and simplicity)	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -
Net Cash from Operations	-€ 1,739.47	€ 3,010.53	€ 3,460.53	€ 3,910.53	€ 4,360.53	€ 4,810.53	€ 5,260.53	€ 5,710.53	€ 6,160.53	€ 6,610.53	€ 7,060.53	€ 7,030.53	€ 55,646.40
Cash Flow from Investing													
Manufacturing Equipment purchase	-€ 150,000.00	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	-€ 150,000.00
Company Phone Purchases	-€ 6,150.00	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	-€ 6,150.00
Investment in R&D (assume capitalized)	-€ 140,000.00	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	-€ 140,000.00
Investment in Patents (assume capitalized)	-€ 7,900.00	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	-€ 7,900.00
Investment in Web Development (assume capitalized)	-€ 7,000.00	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	-€ 7,000.00
Injection Mould Purchase (assume capitalized)	-€ 5,000.00												
Computer/Software purchases	-€ 14,500.00	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	
Cash Flow from Financing													
Seed Investment	€ 300,000.00	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ 300,000.00
Opening Cash Balance	€ -	-€ 32,289.47	-€ 29,278.93	-€ 25,818.40	-€ 21,907.87	-€ 17,547.33	-€ 12,736.80	-€ 7,476.27	-€ 1,765.73	€ 4,394.80	€ 11,005.33	€ 18,065.87	
Ending Cash Balance	-€ 32,289.47	-€ 29,278.93	-€ 25,818.40	-€ 21,907.87	-€ 17,547.33	-€ 12,736.80	-€ 7,476.27	-€ 1,765.73	€ 4,394.80	€ 11,005.33	€ 18,065.87	€ 25,096.40	

Table 6: Sample Statement of Cash Flows FY2027.

Financial Feasibility

Balance Sheet

A snapshot of RideAware's financial position for FY2027 can be seen through the succeeding balance sheet.

As explained previously we have assumed a JIT inventory for optimisation and simplicity purposes, therefore leaving inventory with a 0 balance. Similarly, our accounts receivable balance is also 0 due to the assumption of cash only sales. Detailing the non-current assets section, we have included both tangible and intangible assets.

RideAware Ltd.	
Sample Balance Sheet	
For the Year Ended 31 December 2027	
Assets	
Current Assets	
Cash	€ 25,096.40
Accounts Receivable	€ -
Inventory	€ -
Other Current Assets	€ -
Total Current Assets	€ 25,096.40
Non-Current Assets	
Manufacturing Equipment	€150,000.00
Less: Accumulated Depreciation	-€14,000.00
Computers/Software (e.g., Macbooks)	€ 14,500.00
Less: Accumulated Depreciation	-€ 2,300.00
Employee Phones	€ 6,150.00
Less: Accumulated Depreciation	-€ 1,030.00
Injection Mould	€ 5,000.00
Less: Accumulated Depreciation	-€ 880.00
Intangible Assets	
Prototypes	€140,000.00
Less: Accumulated Amortization	-€ 14,000.00
Website	€ 7,000.00
Less: Accumulated Amortization	-€ 700.00
Patents/Trademarks	€ 7,900.00
Less: Accumulated Amortization	-€ 790.00
Goodwill	€ -
Total Fixed Assets	€296,850.00
Total Assets	€321,946.40
Liabilities	
Current Liabilities	
Accounts Payable	
Income Tax Payable (IE & CHN)	€ 9,935.33
Other Current Liabilities	€ -
Total Current Liabilities	€ 9,935.33
Long-Term Liabilities	
Long-Terrm Notes Payable (Assume none due to budget restriction)	€ -
Other Long-Term Liabilities	€ -
Total Long Term Liabilities	€ -
Total Liabilities	€ 9,935.33
Shareholder's Equity	
Retained Earnings	€ 12,011.08
Seed Investment	€300,000.00
Total Liabilities and Shareholder's Equity	€321,946.40

Table 7: Sample Balance Sheet FY2027.

Tangible assets include line items such as manufacturing equipment, computers/software, employee phones, and injection moulds, all of which are essential to our business operations. These assets were all depreciated accordingly using the straight-line method (see *Appendix 3.1* for details) and netted against their initial costs. Likewise, intangible assets were amortized using the straight-line method (see *Appendix 3.2* for details). Next, our liabilities are minimal due to no additional financing and cash only payments to vendors and suppliers. This leaves our Shareholder's equity section with our initial seed investment and total net income from operations throughout FY2027.

Break Even Analysis

To briefly explain, our break-even analysis sets out 3 cases in a 'best to worst' spectrum in order to account for unforeseen events affecting our financial position in year 1. We have applied a factor of 1.5x to our fixed costs (i.e., base-case/1.5, or base-case*1.5) to model potential outcomes. According to our previous sales forecast (see *concept paper*), we expect to *at least* break even in our base and best cases, while missing the break-even point in our worst case.

RideAware Ltd.			
Break Even Analysis			
For the Year Ended 31 December 2027			
Projected Scenario:	***Base-Case***	***Best-Case***	***Worst-case***
Per Unit Selling Price (rounded)	€ 194.00	€ 194.00	€ 194.00
Per Unit Estimated Cost (rounded)	€ 149.00	€ 149.00	€ 149.00
Contribution Margin	€ 45.00	€ 45.00	€ 45.00
Appoximate Fixed costs	€ 68,053.60	€ 45,369.07	€ 102,080.40
Approximate Break Even Point (units)	1512	1008	2268
Approximate Break Even Point (€ sales)	€ 293,386.63	€ 195,591.09	€ 440,079.95
Projected € Sales in Year 1 (assumed in concept paper calcs.)	€ 388,000.00	€ 388,000.00	€ 388,000.00
Projected Unit Sales in Year 1 (assumed in concept paper calcs.)	2000	2000	2000

Table 8: Sample Break Even Analysis FY2027.

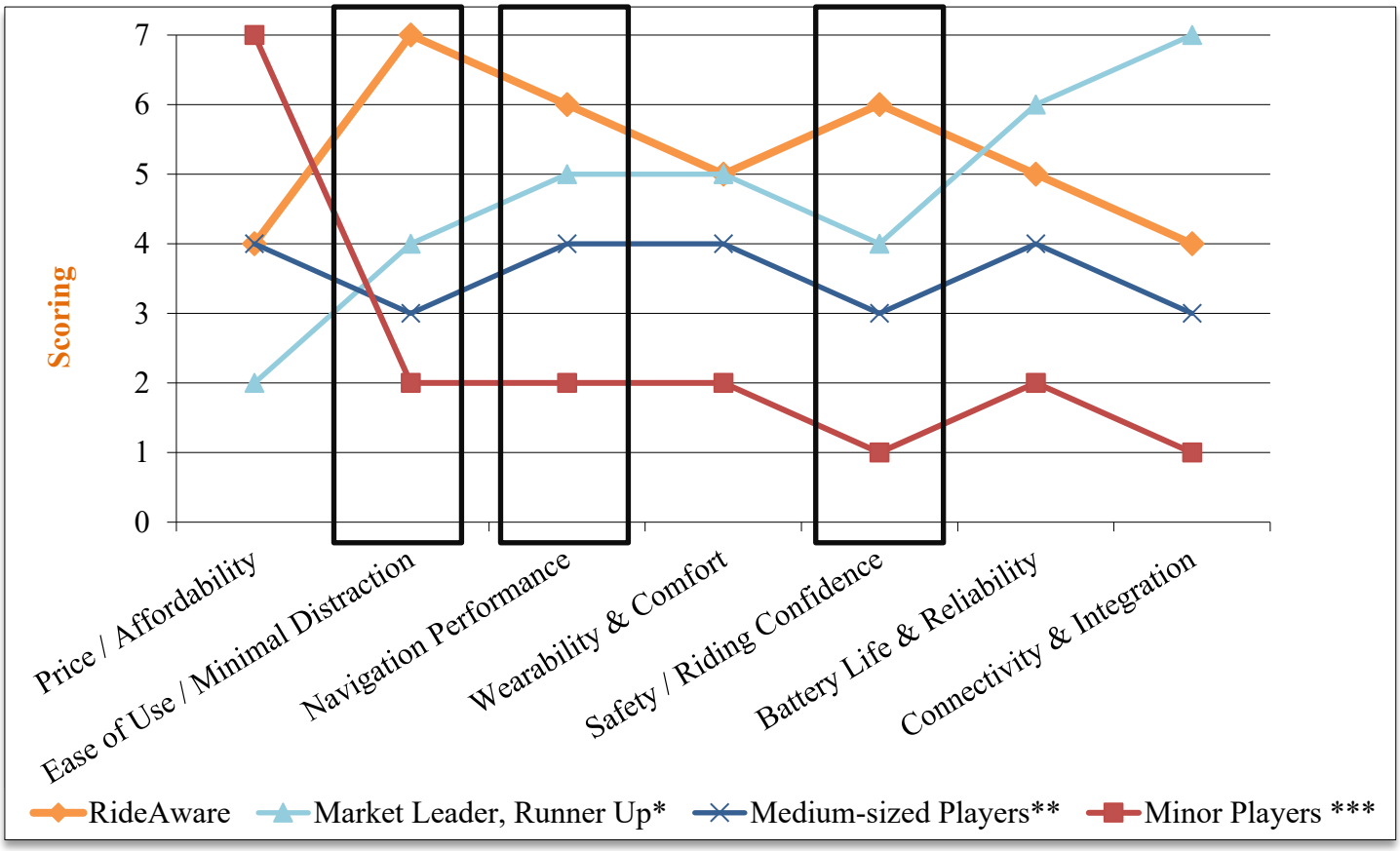
Appendices

Appendix 1: Primary Research Data

Type	Group	Age Group	Location	Number	Riding Purpose	Notes	Name
Interview	Dublin Riders (Younger Group)	20-30	Dublin	3	Commuting, fitness, road cycling	Mature riders and rural/off-road contexts	Anais Wunsch, Erik Gellert, Frankie O'Sullivan
	German Riders (Older Group)	40-60	Germany (urban & rural)	2	Commuting, leisure riding, mountain biking	Purely urban cycling environment	Uwe Helber, Michael Dury
	Mixed Riders (Younger Group)	20-30	Dublin & various European cities	3	Commuting, fitness, road cycling	Combines urban Irish & German cycling contexts	Jakob Schwarze, Joel Helber, Raphael Algöwer

Type	Expert	Area of Expertise	Department at DCU	Main Concerns
Expert Technical Feasibility Survey	Dr. Graham Healy	Brain-Computer Interfaces, Eye-Tracking Interfaces	Engineering and Computing	Power processing and concerns with rear sensor
	Dr. Derek Molloy	Human-Computer-Interaction, Augmented Reality/Virtual Reality	Engineering and Computing	Stated that concept was technically feasible, but commercially infeasible
	Dr. Hyowon Lee	IoT, Edge programming, Embedded Systems, AR/VR	Electronic Engineering	Liability concerns due to risky use-case of product

Appendix 2: Strategy Canvas



* ENGO, Garmin ** Solos, Everysight Raptor and Cosmo Vision *** LUNA Smart Cycling Glasses, Kickstarter, Indiegogo HUD concepts, low-cost OEM smart glasses from Shenzhen

Appendix 3: Financial Feasibility Assumptions

Appendix 3.1: Depreciation Assumptions:

Assumptions for Company Phones (Straight Line Depreciation):			
Useful Life		5	Years
Salvage Value	€	1,000.00	
Initial Cost (Estimated)	€	6,150.00	

Initial Cost Source: Business - Apple (IE). (2025). *iPhone 17 512GB Black*. [online] Available at: https://www.apple.com/ie-business/shop/buy-iphone/iphone-17/6.3-inch-display-512gb-black [Accessed 12 Nov. 2025].

Assumptions for Injection Mould Equipment (Straight Line Depreciation):			
Useful Life		5	Years
Salvage Value	€	600.00	
Initial Cost (Estimated)	€	5,000.00	

Initial Cost Source: See Concept Paper

Assumptions for Macbooks (Straight Line Depreciation):			
Useful Life		5	Years
Salvage Value	€	3,000.00	
Initial Cost (Estimated)	€	14,500.00	

Initial Cost Source: Apple (IE). (n.d.). *Buy 16-inch MacBook Pro*. [online] Available at: https://www.apple.com/ie/shop/buy-mac/macbook-pro/16-inch.

Assumptions for Manufacturing Equipment (Straight Line Depreciation):			
Useful Life		10	Years
Salvage Value	€	10,000.00	
Initial Cost (Estimated)	€	150,000.00	

Initial Cost Source: Wexford Insurance. (2025). *How Much Does It Cost to Start a Manufacturing Business?* [online] Available at: https://www.wexfordins.com/post/how-much-does-it-cost-to-start-a-manufacturing-business [Accessed 12 Nov. 2025].

Appendix 3.2: Amortisation of Intangible Assets:

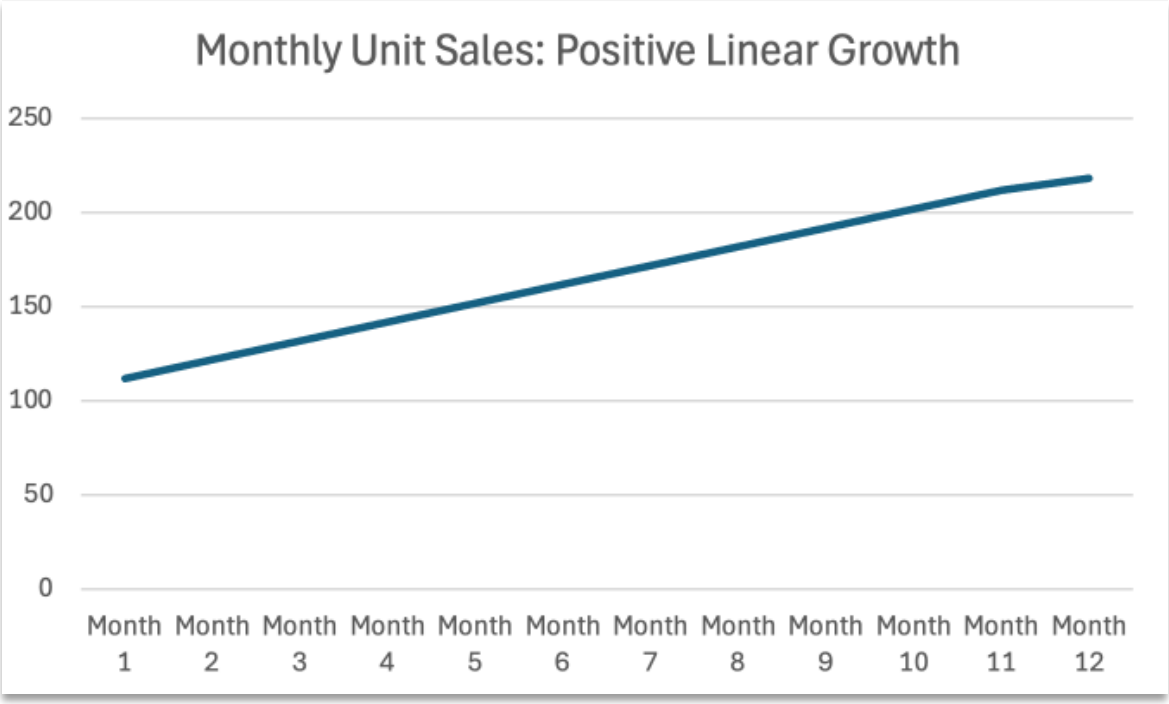
In terms of intangible assets such as prototypes, company website, patents and trademarks amortized on our sample balance sheet, we have reasonably assumed that a useful life of 10 years be applied under the straight-line amortisation approach (Dorland, 2023; Department of Enterprise, Tourism, and Employment, n.d.; Intellectual Property Office of Ireland, n.d.). In addition, given the complexity of estimating the salvage value of intangible assets we have simplified our approach to allow a salvage value of 0.

Appendix 3: Financial Feasibility Assumptions

Appendix 3.3: Monthly Sales Assumptions:

Given the Year 1 Sales Forecast of 2000 units and price per unit of €194 (see *Concept Paper* for details), we can take a simplified approach to calculate the monthly sales. For simplicity we will assume a positive (i.e., increasing) linear growth trend as shown below:

Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9	Month 10	Month 11	Month 12
112	122	132	142	152	162	172	182	192	202	212	218



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DECLARATION OF AUTHORSHIP

Assignment Submission

Student Name(s): Oliver Lopez (46524) , Daniel Cino (46528), Madeleine Helber (45643), Juliana Dury (46562), Aidan Mistry (46510)
Programme: Bachelor of Arts in Global Business
Project Title: New Enterprise Development Project
Module Code: BAA1003
Lecturer: Prof Eric Clinton
Project Due Date: 5 December 2025

We hereby certify that this material which we now submit for assessment on the programme of study leading to the award of Bachelor of Business Studies/ Global Business, is entirely our own work, and has not been taken from the work of others, save, and to the extent that, such work has been cited and acknowledged within the text of our work.

Signed: _____ ID No: (46528) Date: 5.12.2025

Signed: _____ ID No: (46562) Date: 5.12.2025

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